

Capstone Project Briefs

Three end-to-end projects spanning tabular ML, NLP, and computer vision — from the fundamentals of AI through to Convolutional Neural Networks.

EDA & Classical ML → NLP & Deep Learning → CNN & Transfer Learning

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Patient Readmission Risk Predictor

Tabular data · Supervised learning · Healthcare

Tabular / CSV | Binary Classification | Foundational → ML

PROJECT QUESTION

Can you build a machine learning pipeline that predicts whether a diabetic patient will be readmitted to hospital within 30 days of discharge, based on clinical and demographic features — and explain which factors matter most?

CORE SKILLS COVERED

EDA · Feature engineering · Logistic regression ·
Decision trees · Random forest · Evaluation metrics ·
Class imbalance (SMOTE)

DATASET

Diabetes 130-US Hospitals (1999–2008)

101,766 patient records · 50 features · multi-class
readmission label

Kaggle: kaggle.com/datasets/jimschacko/airlines-dataset-to-predict-a-delay

UCI ML Repository: archive.ics.uci.edu/dataset/296/diabetes+130-us+hospitals

KEY DELIVERABLES

- EDA notebook with at least 5 visualisations revealing patterns in the data
- Preprocessing pipeline (missing values, encoding, scaling, imbalance handling with SMOTE)
- Comparison of ≥ 3 models reported with ROC-AUC, F1 score, and confusion matrix
- Feature importance analysis with a written business interpretation
- A short clinical recommendation memo summarising actionable findings

Multilingual Sentiment Analysis on Product Reviews

Text data · NLP · E-commerce

Text / NLP | Multi-class Classification | ML → Deep Learning

PROJECT QUESTION

Can you design a sentiment analysis system that classifies Amazon product reviews (1–5 stars) into positive, neutral, and negative — comparing a traditional TF-IDF + ML baseline against a fine-tuned embedding-based deep learning model?

CORE SKILLS COVERED

Text preprocessing · TF-IDF · Naive Bayes · Word embeddings · LSTM / GRU · Transfer learning basics
· Precision / Recall / F1

DATASET

Amazon US Customer Reviews (Multi-category)

3 M+ reviews across electronics, books & fashion · star rating label

Kaggle: kaggle.com/datasets/cynthiarempel/amazon-us-customer-reviews-dataset

HuggingFace:

huggingface.co/datasets/amazon_us_reviews

KEY DELIVERABLES

- Text cleaning pipeline: tokenisation, stopwords, lemmatisation, label mapping
- Baseline model: TF-IDF + Logistic Regression / Naive Bayes with cross-validation
- Deep model: Embedding layer + LSTM trained from scratch on the same splits
- Side-by-side benchmark table: accuracy, macro F1, and inference time
- Error analysis — what types of reviews fool each model and why?

Real-time Plant Disease Detection with CNN

Image data · Computer vision · Agriculture

Images / Vision | Multi-class Classification | CNN · Transfer Learning

PROJECT QUESTION

Can you build and deploy a convolutional neural network that identifies 38 classes of plant diseases from leaf photographs — and demonstrate that transfer learning with MobileNetV2 outperforms a CNN trained from scratch in both accuracy and training time?

CORE SKILLS COVERED

Image augmentation · CNN architecture · Pooling & convolutions · Transfer learning · Fine-tuning · Grad-CAM · Model deployment (Gradio)

DATASET

PlantVillage Dataset

87,000 labelled leaf images · 38 disease / healthy classes · 256x256 px

Kaggle: kaggle.com/emmarex/plantdisease

GitHub: github.com/spMohanty/PlantVillage-Dataset

KEY DELIVERABLES

- Custom CNN baseline: ≥ 3 conv blocks with batch normalisation and dropout
- Transfer learning model: MobileNetV2 or EfficientNetB0 with fine-tuned top layers
- Training curves (loss & accuracy) and a benchmark comparison table for both models
- Grad-CAM visualisations showing which leaf regions each model attends to
- Simple Gradio or Streamlit demo: upload a leaf photo and receive a disease diagnosis

Note: Each project intentionally builds on the previous one — tabular ML → NLP → computer vision — mirroring the boot camp's progression from classical algorithms to deep learning and CNNs. All datasets are freely available on Kaggle or the sources listed, and are sized to run comfortably on a free Google Colab or Kaggle notebook.